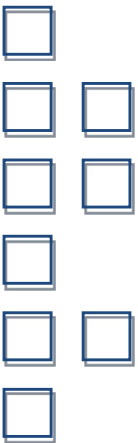


YLM-40-MT FIBER LASER USER GUIDE



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Preface

Please take time to read and understand this User’s Guide in its entirety and familiarize yourself with the operating and maintenance instructions that we have compiled for you before you use the product. We strongly recommend that all operators of the product read and pay particular attention to all safety information contained herein prior to operating the product.

This User’s Guide should stay with the product to provide you and all future operators, users and owners of the product with important operating, safety and other information.

For technical assistance concerning the product, contact IPG Customer Service.

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IPG PHOTONICS CORPORATION

IPG delivers fiber lasers in all domains of laser material processing, medical engineering and scientific institutes and universities. IPG is the accepted market leader in this modern fiber laser technology. The IPG Photonics Corporation contains subsidiaries in many European, Asiatic and American developed countries. Production sites for these lasers are in both Oxford, Massachusetts in the US and in Burbach, Germany with a vertical range of manufacture of 90%. This helps to guarantee a high grade of competence and reliability.

High output power, excellent beam quality, high stability and mobility are outstanding predicates of the fiber laser. The superiority of the fiber laser is efficiency, flexibility, modularity, thermal and mechanical robustness that comes along with a simple constructive assembly. Further important characteristics for the industrial use are low running costs.

Economic feasibility studies show, that it is possible to realize faster, better and more cost-effective industrial applications with high power fiber laser. IPG has the production capacity for a production in high piece numbers. Additional information can be found at www.ipgphotonics.com.

IPG Companies with present service



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INTRODUCTION

The YL-Series are high power fiber lasers which meet many customer requirements. These lasers are flexible, ready-to-use instruments allowing them to replace bulky and less efficient lasers. Direct pumping by high quality laser diodes operating around 970 nm is used in all models. IPG Photonics produces these lasers in three versions, modular, bench-top and rack mount.

The IPG YLM-40-MT laser is a Class 4 laser product and is designed and tested with safety in mind. By following this User's Guide and applying sound laser safety practices, it will be a safe and reliable device.

Laser light exhibits unique characteristics that pose safety hazards that are not normally associated with other light sources. Therefore, all operators and people near the laser must be aware of these special hazards.

In order to ensure the safe operation and optimal performance of the product, please follow these warnings in addition to the other information contained elsewhere in this manual. These safety precautions must be observed during all phases of operation, maintenance, and service of this instrument. Operators are urged to adhere to these recommendations and to apply sound laser safety practices at all times. Never open the device. There are no user serviceable parts (internally), equipment or assemblies associated with this product. All internal service and maintenance will be performed only by qualified IPG personnel.

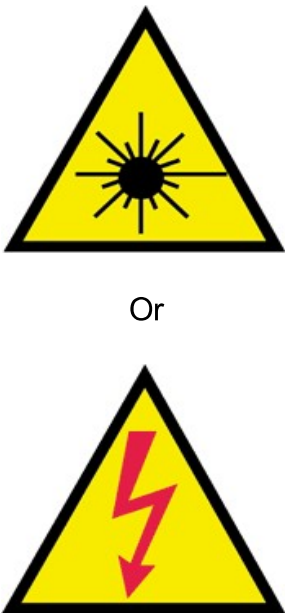
This product is covered by the U.S. Pat. Nos. US7773655 and US 8711894, Patent Publications: US2017/0018906 and any foreign counterparts thereof, and other patents pending.

SAFETY INFORMATION

Safety Conventions

This User Guide uses various words and symbols (Table 1) are designed to call your attention to hazards or important information.

Table 1: User Guide Safety Symbols

SYMBOL	DESCRIPTION
 Or	WARNING: Refers to a potential <i>personal</i> hazard. ( Electrical) ( Laser radiation) It requires a procedure that, if not correctly followed, may result in bodily harm to you and/or others. Do not proceed beyond the WARNING sign until you completely understand and meet the required conditions.

Safety Features and Compliance to Government Requirements

Compliance to Regulatory Standards (if applicable)

This product is specifically designed for integration into other equipment and is therefore considered a non-compliant, OEM component. It is the OEM's responsibility to ensure compliance with all government requirements with respect to this product.

Electromagnetic Compatibility

This product is designed to meet EMC requirements. However, it is the OEM's responsibility to ensure compliance with all EMC requirements with respect to the product.

Laser Classification

The governmental standards and requirements specify that lasers must be classified according to their output power or energy and the laser wavelength. YLM-40-MT lasers are classified as a high power Class 4 laser instrument under 21 CFR, 1040.10. According to the European Community standards, this device is classified as Class 4 based on EN 60825-1.

This product emits invisible laser radiation at two independent channels. One channel emits invisible laser radiation at or around a wavelength of **975nm**, and the total light power radiated from the optical output is greater than **18 watts CW**. The second channel emits invisible laser radiation at or around a wavelength of **1064 nm**, and the total light power radiated from the optical output is greater than **46 watts CW**. This product is also equipped with a Class 2M visible laser operating at a wavelength at or around **658nm** and the total light radiated from this visible laser is **<1mW CW**. Laser emission of each channel and the visible laser is delivered via a single fiber optic delivery cable terminated with an SMA connector.

Direct or indirect exposure of this level of light intensity may cause damage to the eye or skin. Despite the radiation being invisible, the beam may cause irreversible damage to the retina and/or cornea. Appropriate and approved laser safety eyewear must be worn at all times while the laser is operational. Warning: You must use appropriate laser safety eyewear when operating this device. The selection of appropriate laser safety eyewear requires that the end user accurately identify the range of wavelengths emitted from this product.

The end user must ensure that the laser safety eyewear used protects against light emitted by the device over its entire range of wavelengths. Please review the safety labeling on the product (Figure 1: Safety Label Locations) and verify that the personal protective equipment (e.g. enclosures, viewing windows or viewports, eyewear, etc.) being utilized is adequate for the output power and wavelength ranges listed on the product.

Suppliers include LaserVision USA, Kentek Corporation and Rockwell Laser Industries offer this laser safety material and/or equipment. There are other laser personal protective equipment providers. IPG provides the names of these providers solely as a convenience and does not endorse or recommend any of them, or their products or services. Furthermore, IPG assumes no liability for any of their recommendations, products or services.

Whether the laser is used in a new installation or to retrofit an existing system, the end user is solely responsible for determining the suitability of all personal protective equipment.

⚠ Caution: Do not install or terminate the SMA connector when laser is active.

Laser Control

The laser cannot be turned on or operated until first providing power and Lemo connections (Refer to the Electrical Power Connections section starting on page 15).

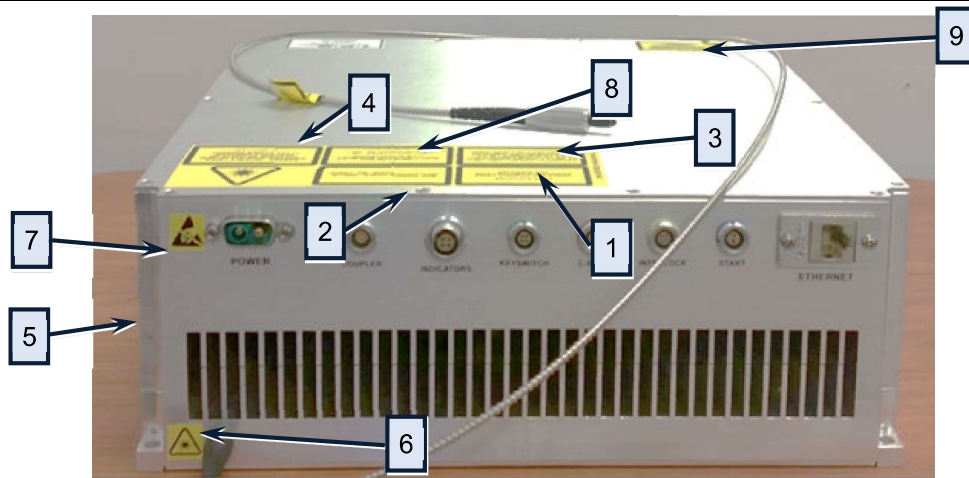
⚠ Caution: *Use of controls or adjustments, or performance of procedures other than those specified herein, may result in hazardous radiation exposure.*

⚠ Caution: Use of the system in a manner other than that described herein may impair the protection provided by the system.

Safety Label Locations

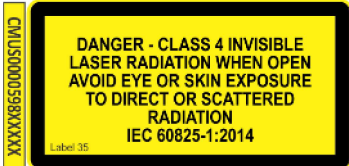
Figure 1: Safety Label Locations show the required laser safety labels and the locations. These include warning labels indicating removable or displaceable protective housings, apertures through which laser radiation is emitted and labels of certification and identification.

Figure 1: Safety Label Locations



Model YLM-40-MT (YLMXM040C106401X) shown

1. Class 4 Laser Product Top Cover of Laser Module (CMUS2001246XXXXXU)	2. Certification (for US product) Top Cover of Laser Module	3. Class 2M Laser Product Label for Guide Laser Top Cover of Laser Module
4. Aperture Label Top Cover of Laser Module, Output SMA	5. Identification Plate (Products Made in the United States) Side Of Module	6. Laser Radiation Hazard Label Top Cover of Laser Module Output SMA, Fiber Exit

		
7. ESD Hazard Label ¹	8. Warning Logotype	9. Class 4 Laser Product, when opened Top and Bottom Cover of Laser Module

General Safety Instructions

⚠ Warning: You must exercise caution to avoid/minimize specular reflections as these reflections occur at the laser's wavelength and are invisible!

Specular Reflections

Often there can be numerous secondary laser beams produced at various angles near the laser. These beams are called specular reflections and are produced when the laser reflects off a smooth surface of the primary beam. Although these secondary beams may be less powerful than the total power emitted from the laser, the intensity may be great enough to cause damage to the eyes and skin as well as materials surrounding the laser.

Equipment and Solvents

Light-sensitive elements in equipment, such as video cameras, photomultipliers and photodiodes may also become damaged from exposure to the laser light.

⚠ Caution: The laser light is strong enough to cut or weld metal, burn skin, clothing and paint. In addition, this light can ignite volatile substances such as alcohol, gasoline, ether and other solvents.

Exposure to solvents or other flammable materials and gases must be avoided and must be relocated away from this device.

Safety Recommendations

IPG recommends that you follow these procedures to operate the IPG laser safely:

- Never look directly into the laser output when power is supplied to the laser.
- Avoid positioning the laser and all optical components at eye level.
- Provide enclosures for laser beam.
- Ensure that all personal protective equipment (PPE) is suitable for the output power and wavelength range listed on the laser safety labels that are affixed to the product.

¹ Electrostatic Hazard (ESD): Indicates an ESD sensitive device. Electrostatic discharge can change the electrical characteristics of a semiconductor device, degrading or destroying it. Electrostatic discharge also may upset the normal operation of an electronic system, causing equipment malfunction or failure.

- Use the laser in a room with access controlled by door interlocks. Post warning signs. Limit access to the area to individuals who are trained in laser safety while operating the laser.
- Avoid using the laser in a darkened environment.
- Always switch the laser off when working with the output such as mounting the SMA into a fixture, etc. If necessary, align the output at low output power and then increase the output power gradually.
- Do not install or terminate the SMA when laser is active.
- If this instrument is used in a manner not specified in this document, the protection provided by the instrument may be impaired and the warranty will be voided.

Optical Safety

⚠ Caution: If the output of the device is delivered through a lens with an anti-reflection coating, make sure that the lens is of good quality and clean. For cleaning instructions, refer to the “Optical SMA Inspection and Cleaning Guide”.

Any dust on the end of the SMA assembly can burn the output and damage the laser.

Electrical Safety

⚠ Warning: The input voltage to the laser is potentially lethal. All electrical cables and connections should be treated as if it were a harmful level. All parts of the electrical cable, connector or device housing should be considered dangerous.

1. Make sure this instrument is properly grounded through the protective conductor of the DC power cable. Any interruption of the protective grounding conductor from the protective earth terminal can result in personal injury.
2. Always use your device in conjunction with properly grounded power source.
3. For continued protection against fire hazard, replace the line fuses (if applicable) with only the same types and ratings. The use of other fuses or material is prohibited.
4. Before supplying the power to the instrument, make sure that the correct voltage of the DC power source is used. Failure to use the correct voltage could cause damage to the instrument.
5. Before switching the power on, ensure the voltage corresponds to the specified level.
6. There are no operator serviceable parts inside. Refer all servicing to qualified IPG personnel. To prevent electrical shock, do not remove covers. Any tampering with the product will void the warranty.

Environmental Safety

⚠ Warning: Never look directly into a live fiber or collimator and make sure that you wear appropriate laser safety eyewear at all times while operating the product.

Proper enclosures must be used to secure a laser safe work area. This includes but is not limited to laser safety signs, interlocks, appropriate warning devices and training/safety procedures. In addition, it is important to install the output assembly away from eye level.

⚠ Warning: Ensure that all personal protective equipment (PPE) is suitable for the output power and wavelength range listed on the laser safety labels that are affixed to the product.

The interaction between the laser and the material being processed can also generate high intensity UV and visible (collateral) radiation. Ensure that all laser enclosures are in place to prevent injury from visible radiation.

⚠ Caution: Damage to the laser is possible, unless caution is employed in operating the device.

IPG provides the following recommendations to promote the long life of the IPG laser:

- Do not expose the device to a high moisture environment (>90% humidity).
- Make sure there is sufficient airflow to cool the device, any objects or debris that cover the ventilation holes must be removed at all times.
- Operation at higher temperatures will accelerate aging, increase threshold current and lower slope efficiency. If the device is overheated, do not use it and call IPG for assistance.

- Ensure that the work area is properly ventilated. Gases, sparks and debris that can be generated from interaction between the laser and the work surface can pose additional safety hazards.
- It is recommended that filter media (not provided) be used in dirty environments.

Additional Safety Resources

For additional information regarding Laser Safety, please refer to the list below:

Laser Institute of America (LIA)

13501 Ingenuity Drive, Suite 128

Orlando, Florida 32826

Phone: 407.380.1553, Fax: 407.380.5588

Toll Free: 1.800.34.LASER

American National Standards Institute

ANSI Z136.1, American National Standard for the Safe Use of Lasers

(Available through LIA)

International Electro-technical Commission

IEC 60825-1

Safety of laser products –

Part 1:

Equipment classification, requirements and user's guide.

(Available through LIA)

Center for Devices and Radiological Health

21 CFR 1040.10 – Performance Standards for Light-Emitting Products

US Department of Labor – OSHA

Publication 8-1.7 – Guidelines for Laser Safety and Hazard Assessment.

Laser Safety Equipment

Laurin Publishing

Laser safety equipment and Buyer's Guides

IPG Photonics recommends that the user of this product investigate any local, state or country requirements as well as facility or building requirements that may apply to installing or using a laser or laser system.

PRODUCT DESCRIPTION

The YLM-40-MT Fiber Laser is a custom OEM laser, designed specifically for the needs of our customer. This product is specifically developed to be incorporated as a component level item and is to be integrated into their end use application.

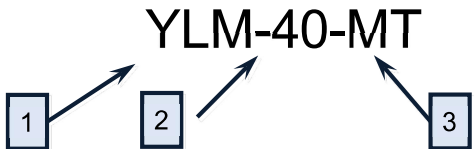
Main Features:

- High quality fiber output
- High power
- Reliable, long lifetime
- Compact, rugged package
- Efficient
- External computer interface

Certification

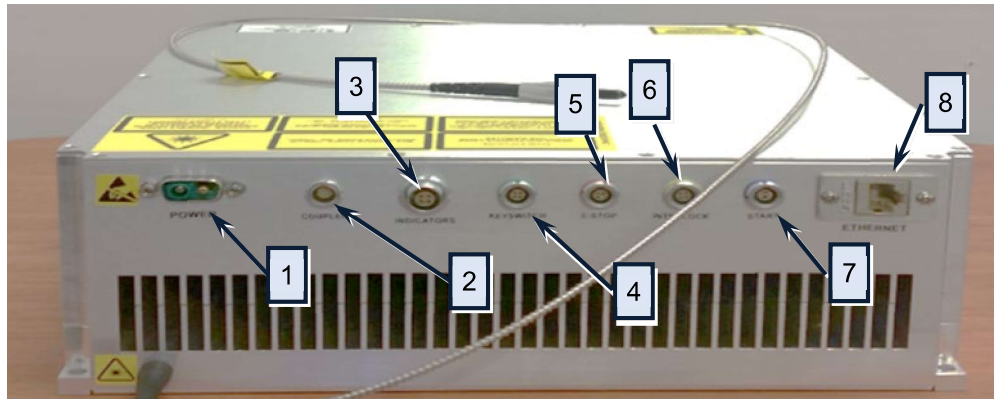
IPG certifies that this instrument has been thoroughly tested and inspected, and found to meet published specifications prior to shipping. Upon receipt of the device, check the packaging and parts for any possible damage that may have occurred in transit. If damage is apparent, please contact IPG immediately.

Table 2: Laser Designation Codes

	
ITEM	CODE
1. "YLM"	<u>Y</u> tterbium <u>L</u> aser <u>M</u> odule
2. Power in watts	40- nominal (CW) power
3. Additional Information	Characters denoting customer: Medtronic

Front Side View

Figure 2: Front Side of Module Layout



Model YLM-40-MT (YLMXM040C106401X) shown

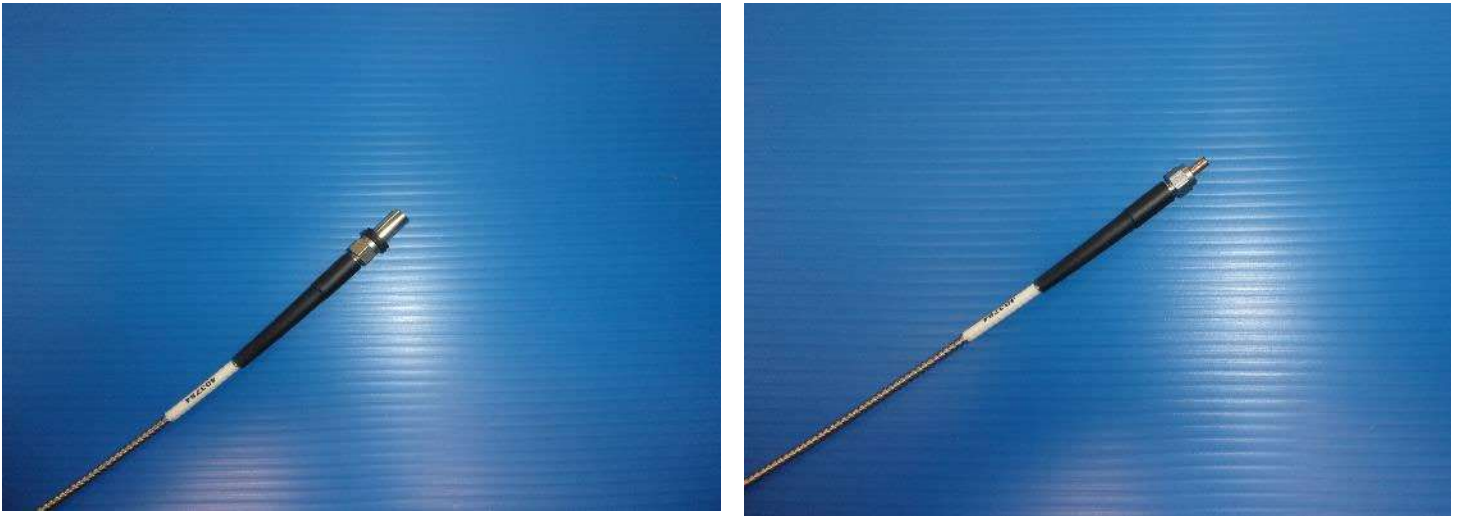
ITEM	DESCRIPTION
1. DB-2W2	Main power supply.
2. Coupler Lemo Interface Connector	Connection for Coupler interface cable.
3. Indicator Lemo Interface Connector	Connection for Indicator interface cable.
4. Key switch Lemo Interface Connector	Connection for Key switch interface cable.
5. Emergency Stop Lemo Interface Connector	Connection for Emergency Stop interface cable.
6. Interlock Lemo Interface Connector	Connection for Interlock interface cable.
7. Start Lemo Interface Connector	Connection for Start interface cable.
8. Ethernet Connection	Connection for Ethernet cable.

Detailed information on the Lemo connections is located in the “Interfaces and Commands” section.

Modules with a SMA

SMA's do not have a protective window on the end of the connector. The fiber tip is **exposed** when the cap is off; therefore, care **MUST** be taken when working with the SMA when the cap is off. The SMA end cap must be removed prior to use and should be used when storing the module. Cleaning of the SMA should be performed as needed using the same materials and techniques described in the "Optical SMA Inspection and Cleaning Guide" section.

Figure 3: SMA



SPECIFICATIONS

Please refer to Medtronic product specification for details (M704430B323PS) and any attached document(s) it references.

UNPACKING INSTRUCTIONS

Unpacking Procedure

All laser modules that are packaged in foam insulated cardboard boxes. In order to minimize the risk of damage to your device, IPG recommends that you unpack your laser using the following procedure.

Figure 4: Packaging Contents



If the packaging shows any signs of external damage, check unit for damages and notify the shipping agent immediately.

Particular care must be taken when removing the unit from the packing case to ensure that the fiber optic cable is not broken or damaged. A comprehensive packing list is included with the system documentation. Upon receipt of the laser, check all items against this list and contact IPG immediately if any of the items are missing or if any damage to the unit is evident. If any damage to the unit is evident or suspected, do not attempt to install or operate the laser in any case.

Remove all contents of the packaging case, taking care to retain all packaging materials and inserts. Particular care must be taken when removing the unit from the packing case to ensure that the fiber optic cable is not broken and damaged. A comprehensive packing list is included with the system documentation. Upon receipt of the laser, check all items against this list and contact IPG immediately if any of the items are missing or if any damage to the unit is evident. If any damage to the unit is evident or suspected, do not attempt to install or operate the laser in any case.

Retain all packaging for future transportation or storage needs.

USING YOUR DEVICE

Precautions

⚠ Caution: Refer to the Medtronic product specification for details (M704430B323PS) for proper electrical power requirements.

Before switching the power on, make sure that the incoming DC voltages are equal to the level noted in the specification.

Operate only with sufficient air flow for fans to cool the module during operation.

Allow ample amount of time for the fans to cool the module. Once the module has been cooled, de-energize the power supply and then the housekeeping supply.

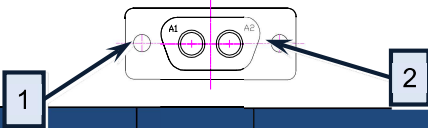
Electrical Power Connections

For the electrical specification, refer to the Medtronic product specification for details (M704430B323PS) and any attached document(s) it references.

Powering the Module

The module is powered through the 2W2 power connector. It is designed to operate with a standard off-the-shelf 24VDC power supply.

Figure 5: 2W2 2-Pin Power Supply Pin Assignment



Pin #	Signal Name	Level	Type	Function
A1	Supply Return	0V		Supply return (pump diodes feed)
A2	Main Power Supply	Up	Input	+ VDC supply (pump diodes feed)

External Interfaces

The interface is provided using a series of Lemo connectors (Figure 2: Front Side of Module Layout). Refer to the “Interfaces and Commands” section for setting up these interfaces.

Interface Wire Specification

The minimum wire gage is 18AWG at 15 meters (30 meters maximum regardless of gauge). The gage of the wire must increase as the distance increases. For connectivity, the wiring and/or cabling must have an overall shield to ensure proper functionality. The shield is to cover over all conductors and terminate at the unit where the conductors enter/exit the unit.

Operation Modes

YLM-40-MT model will operate in **CW** mode only.

INTERFACES AND COMMANDS

Customer Connections

Any of the connections that use the same connector will be color coded. The following tables provide a summary of the laser's connections, mating parts and color coding.

Table 3: Customer Connections: Laser Designation Codes

Function	Connector P/N Color Washer	# of Pins	Size	Color	Mate / Color Strain Relief
X-1: Main 24V Power	EEG.0B.304.CYM/GRA.0S.269.GR	4	10mm	Red	FGG.0B.304.CYCZ/GMA.0B.0XX.RR
X-9: Keyswitch	EEG.0B.304.CYM/GRA.0S.269.GV	4	10mm	Green	FGG.0B.304.CYCZ/GMA.0B.0XX.RV
X-5: Start Button	EEG.0B.302.CYM/GRA.0S.269.GA	2	10mm	Blue	FGG.0B.302.CYCZ/GMA.0B.0XX.RA
X-8: Indicators	EEG.1B.304.CYM/NONE	4	14mm	Grey	FGG.1B.304.CYCZ/GMA.1B.040.RG
X-10: External Interlock	EEG.0B.304.CYM/GRA.0S.269.GJ	4	10mm	Yellow	FGG.0B.304.CYCZ/GMA.0B.0XX.RJ
X-11: E-Stop	EEG.0B.304.CYM/GRA.0S.269.GR	4	10mm	Red	FGG.0B.304.CYCZ/GMA.0B.0XX.RR
X-6: Coupler	EEG.0B.309.CYM/NONE	9	10mm	Gray	FGG-0B-309-CYCD52Z /GMA.0B.045.DG

Main Power: The main power connection provides both the housekeeping and laser diode power.

- 2W2CP DSUB Male (populated with 2 PINS)
- Mating Part 2W2C Female (populated with 2 sockets)
- Example: Connect 302W2CSXX99A10X with 132A11019X Sockets

Table 4: Customer Connections: X1 Main Power

POS	Name	Description
A1	RTN	Return to +24V
A2	+24V	+24V Input

Key switch: The key switch interface is operated by connecting a dry contact between the two pins of each loop. Closing these connections completes the interlock loop. When this connection is open, the HK power to the module is still present allowing communication to the laser, however it is not possible to turn on the laser diode power.

Table 5: Customer Connections: X-9 Keyswitch

POS	Name	Description
1	KS1A	Key switch loop 1, connect to KS1B to close
2	KS1B	Key switch loop 1, connect to KS1A to close
3	KS2A	Key switch loop 2, connect to KS2B to close
4	KS2B	Key switch loop 2, connect to KS2A to close

Start Button: The start button interface is operated by momentarily closing a dry contact between the two pins. If the interlocks are satisfied, activation of this signal will turn on the pass switches and the main power will be connected the laser diodes.

Table 6: Customer Connections: X-5 Start Button

POS	Name	Description
1	Start (+)	Start (+) potential terminal. Temporarily connect to Start (-) to enable the laser diode power.
2	Start (-)	Start (-) potential terminal.

Indicators: Each indicator output is a 24V ,100mA current limited output.

Table 7: Customer Connections: X-8 Indicators

<i>POS</i>	<i>Name</i>	<i>Description</i>
1	GND	All indicator signals referenced to this connection.
2	ERROR	+24V output when activated.
3	EMON	+24V output when activated.
4	READY	+24V output when activated.

External Interlock: There are two interlock loops. Both of the loops are required to be closed before the pass switches can be activated. A dry contact is placed across the channel's A and B connections in order to close it.

Table 8: Customer Connections: X-10 External Interlock

<i>POS</i>	<i>Name</i>	<i>Description</i>
1	CH1 A	CH1 Loop, connect to CH1 B to close.
2	CH1 B	CH1 Loop, connect to CH1 A to close.
3	CH2 A	CH2 Loop, connect to CH2 B to close.
4	CH2 B	CH2 Loop, connect to CH2 A to close

E-Stop: There are two interlock loops. Both of the loops are required to be closed before the pass switches can be activated. A dry contact is placed across the channel's A and B connections in order to close it.

Table 9: Customer Connections: X-11 E-STOP

<i>POS</i>	<i>Name</i>	<i>Description</i>
1	CH1 A	CH1 Loop, connect to CH1 B to close.
2	CH1 B	CH1 Loop, connect to CH1 A to close.
3	CH2 A	CH2 Loop, connect to CH2 B to close.
4	CH2 B	CH2 Loop, connect to CH2 A to close

Coupler: There are four interlock switches and an output power monitor photodiode connection available for the output fiber coupler.

Table 10: Customer Connections: X-6 Coupler

<i>POS</i>	<i>Name</i>	<i>Description</i>
1	OPTILK1_A	Fiber interlock Ch1
2	OPTILK1_B	Fiber interlock Ch1
3	OPTILK2_A	Fiber interlock Ch2
4	OPTILK2_B	Fiber interlock Ch2
5	PD4-A	PD4 Anode
6	PD4-K	PD4 Cathode
7	RESERVED	RESERVED
8	RESERVED	RESERVED
9	RESERVED	---

Ethernet TCP/IP Interface

The IP address is 192.168.59.70. The IP address of the laser is 192.168.59.70. The laser will listen for connections on port 10001. The command must be sent as a single string in a single packet. The individual commands are described in the “Interface Commands” section of the manual. TCP timeout socket is 30 seconds.

Table 11: Ethernet Interface Pinout

PIN	DESCRIPTION	NOTES
1	TX+	Transmit Data +
2	TX-	Transmit Data -
3	RX+	Receive Data +
4	N/C	Not Connected
5	N/C	Not Connected
6	RX-	Receive Data -
7	N/C	Not Connected
8	N/C	Not Connected

Interface Commands

All commands and responses will consist of printable ASCII characters. Commands are three to five letter mnemonic codes followed by a parameter, if required. All commands and responses will be terminated with a <Carriage Return> (CR, 0x0D, \r) character. If a CR terminated string is received, but no valid command is found, a response of “BCMD” will be sent. The commands are shown here as all uppercase for clarity; the actual commands are not case sensitive. A space character is also shown between the command and parameter for clarity; the space is not required.

Every command will generate a response. The responses generally consist of the command echoed back. If there is a returned value, it will be separated from the echoed command by a ‘:’ character.

Table 12: Interface Commands

Code	Description			Example
ABF	Aiming Beam OFF			Sent: "ABF" Response: "ABF"
ABN	Aiming Beam ON			Sent: "ABN" Response: "ABN"
DDHCP	Disable DHCP – Disables DHCP client.			Sent: "DDHCP" Response: "DDHCP"
DWDT	Disables the watchdog timer.			Sent: "DWDT" Response: "DWDT"
EDHCP	Enable DHCP – Enables DHCP client.			Sent: "EDHCP" Response: "EDHCP"
EMOFF	Stop Emission – Stops emission.			Sent: "EMOFF" Response: "EMOFF"
EMON	Start Emission – Starts emission.			Sent: "EMON" Response: "EMON"
EWDT	Enables the watchdog timer.			Sent: "EWDT" Response: "EWDT"
HELP	In case of no parameters returns the list of applicable commands. In case of a specified command name as a parameter returns the command description.			---
RBP	Reads the current output power at the cart's bulkhead SMA connector.			Sent: "RBP" Response: "RBP 26.8"
RBT	Read Board Temperature – Reads the temperature of the laser's control board. The response will be an echo of the command, a delimiter of ":", and the temperature in degrees centigrade.			Sent: "RBT" Response: "RBT 36.6"
RCE	Reset Critical Error - Followed by the code received from IPG clears critical errors.			Sent: "RCE 1123456123" Response: "RCE" Sent: "RCE 034" Response: "ERR: Code Incorrect"
RCS	Read Current Setpoint - Reads the setpoint for the LD current. The response is the command echoed back, followed by a delimiter of ":", and then the current setpoint in %.	Beam ID	Description	Sent: "RCS 1" Response: "RCS 1: 56.7" (indicates that the LD current setpoint is 56.7% for beam ID 1)
		0	970 nm beam	
		1	1064 nm beam	
RDC	Reads the diode current of the selected beam.	Beam ID	Description	Sent: "RDC 1" Response: "RDC 1: 56.7" (indicates that the LD current setpoint is 56.7% for beam ID 1)
		0	970 nm beam	
		1	1064 nm beam	
RDGW	Read Default Gateway - Reads the current default gateway of the device. The response is the command echoed back, followed by a delimiter of ":", and then the default gateway in dot-decimal notation.			Sent: "RDGW" Response: "RDGW: 192.168.0.1"
RDHCP	Read DHCP – Reads the current status of DHCP client function. The response is the command echoed back, followed by a delimiter of ":", and then either "ON" or "OFF".			Sent: "RDHCP" Response: "RDHCP: Off"
REC	Read Error Counter – Reads critical error counter.			Sent: "REC" Response: "REC: 37"
RERR	Reset Errors – Resets any resettable errors.			Sent: "RERR" Response: "RERR"
RET	Read Elapsed Time – Reads the elapsed time the laser has been ON. The time is returned in minutes.	Beam ID	Description	Sent: "RET" Response: "RET: 1105"
		0	970 nm beam	
		1	1064 nm beam	
RFV	Read current software revision.			Sent: "RFV" Response: "RFV: 1.0.147 "
RICDT	Read Input Circuit Discharge time - Reads the fall time of the main power supply voltage. The units are in milliseconds (ms).			Sent: "RICDT" Response: "RICDT: 25"
RIP	Read IP – Reads the current IP address of the device. The response is the command echoed back, followed by a delimiter of ":", and then the IP address in dot-decimal notation.			Sent: "RIP" Response: "RIP: 192.168.59.70"
RIS	Returns the status of all interlocks.	Beam ID	Description	Sent: "RIS" Response: "RIS: 2" E-stop interlock is open. Sent: "RIS" Response: "RIS: 0" All interlocks are closed.
		0	Fiber Proximity, Interlock Open	
		1	E-Stop, Interlock Open	
		2	External, Interlock Open	
		3	Key Switch, Interlock Open	
		4-31	Reserved	
RLET	Returns the length of time that the selected beam was emitting the last time it was turned on in seconds.	Beam ID	Description	Sent: "RLET" Response: "RLET 1: 263123"
		0	970nm beam	
		1	1064nm beam	

Code	Description			Example
RLHN	Read Local Host Name – Reads the current local host name of the device. The response is the command echoed back, followed by a delimiter of “:”, and then the name.			Sent: “RLHN” Response: “RLHN: IPG-12004020”
RMAC	Read MAC Address – Reads the current MAC address of the laser. The response is the command echoed back, followed by a delimiter of “:”, and then the MAC address in a form of six groups of two hexadecimal digits, separated by hyphens (-).			Sent: “RMAC” Response: “RMAC: A1-B2-C3-D4-E5-F6”
RMAN	Returns the name of the manufacturer of the device.			Sent: “RMAN” Response: “RMAN: IPG Photonics
RMASK	Read Subnet Mask – Reads the current subnet mask of the device. The response is the command echoed back, followed by a delimiter of “:”, and then the subnet mask in dot-decimal notation.			Sent: “RMASK” Response: “RMASK: 255.255.0.0”
RMEC	Read Module Error Code - Returns error code stored in the laser or zero if normal operation.			Sent: “RMEC” Response: “RMEC: 0”
RMN	Read Model Name – Reads the model name of the device.			Sent: “RMN” Response: “RMN: YLM-40-MT”
RNC	Read Minimum Current Setpoint – Reads the minimum current setpoint that can be set in the laser. The response will be the command echoed back, followed by a delimiter of “:”, then the minimum current as a percentage of the maximum.	Beam ID	Description	Sent: “RNC 0” Response: “RNC: 10.0” (indicates that the minimum setpoint is 10.0 % for beam ID 0)
		0	970 nm beam	
		1	1064 nm beam	
ROP	Read Output Power – Reads the output power in watts. The response will be the command echoed back, a delimiter of “:”, and then either the power in watts, “Off” if the emission is off, or “Low” if the power is below the reliable measurement threshold of the laser.	Sent: “ROP” Response: “ROP: 96” (indicates that the output power is 96 watts)		
		Sent: “ROP” Response: “ROP: Off” (indicates that emission is off)		
		Sent: “ROP” Response: “ROP: Low” (indicates that the output power is below the accurate measurement range of the laser)		
RSN	Read Serial Number – Reads the serial number of the device.			Sent: “RSN” Response: “RSN: 6103081”
RSTIP	Reset TCP/IP Settings – Resets the settings to the default ones:	Setting	Default Value	Sent: “RSTIP” Response: “RSTIP”
		DHCP Client	Disabled	
		IP Address	192.168.59.70	
		Default Gateway	192.168.59.1	
		Subnet Mask	255.255.255.0	
		DHCP Client	Disabled	
RSWB	Returns the ID of the currently selected working beam.	Beam ID	Description	Sent: “RSWB” Response: “RSWB 1” (indicates that the selected beam ID is 1)
		0	970 nm beam	
		1	1064 nm beam	
RWDT	Resets the watchdog timer.			Sent: “RWDT” Response: “RWDT
SDC	Set Diode Current – Sets the diode current. The units are in percent of maximum current. The setpoint must be below 100% and above the minimum current setpoint. The current may also be set to 0. The response from the laser will be the command echoed back, a delimiter of “:”, and then the current setpoint for the laser. A value that is outside the acceptable range will receive a response of “ERR: Out of Range”.	Beam ID	Description	Sent: “SDC 1 34.2” Response: “SDC 1: 34.2” (current Setpoint is set to 34.2% for beam ID 1) Sent: “SDC 1 104.2” Response: “ERR: Out of Range” (the setpoint is unchanged)
		0	970 nm beam	
		1	1064 nm beam	
SDGW	Set Default Gateway - Followed by a number in dot-decimal notation sets the default gateway for the laser.			Sent: “SDGW 192.168.0.1” Response: “SDGW: 192.168.0.1”
SIP	Set IP – Followed by a number in dot-decimal notation sets the IP address for the laser.			Sent: “SIP 192.168.9.215” Response: “SIP: 192.168.9.215“
SLHN	Set Local Host Name - Specifies the name of the device within the network. The maximum number of characters for the name is 15.			Sent: “SLHN IPG-12004020” Response: “SLHN: IPG-12004020”
SMAC	Set MAC Address – Followed by six groups of two hexadecimal digits, separated by hyphens (-), sets the MAC address for the laser.			Sent: “SMAC 12-34-56-78-EF-EF” Response: “SMAC: 12-34-56-78-EF-EF”
SMASK	Set Subnet Mask – Followed by a number in dot-decimal notation sets the subnet mask for the laser.			Sent: “SMASK 255.255.0.0” Response: “SMASK: 255.255.0.0 “

Code	Description			Example
STA	Read device status – The status is reported as a bit-encoded 32-bit word. Undefined bits or bits defined as “Reserved” can be in any state and should be ignored. Each of the bits have the following meaning:	Bit	Meaning if “0”	Meaning if “1”
		1	Normal Operation	Overheat
		2	All Working Beam Emission Off	Any Working Beam Emission On
		3	Back Reflection OK	High Back Reflection
		4	Laser not ready	Laser in ready mode
		5	Reserved (MNav)	Reserved (MNav)
		6	Reserved (IPG)	Reserved (IPG)
		7	Watchdog disabled	Watchdog enabled
		8	Aiming Beam Off	Aiming Beam On
		9	Reserved (MNav)	Reserved (MNav)
		10	Reserved (MNav)	Reserved (MNav)
		11	Power Supply OK	Power Supply Out of Range
		12	Reserved (MNav)	Reserved (MNav)
		13	Reserved (MNav)	Reserved (MNav)
		14	Normal Operation	External Shutdown
		15	Normal Operation	Power Sensor Warning
		16	Reserved (MNav)	Reserved (MNav)
		17	Reserved (MNav)	Reserved (MNav)
		18	Reserved (MNav)	Reserved (MNav)
		19	Normal Operation	Power Supply Failure
		20	Reserved (IPG)	Reserved (IPG)
		21	Reserved (IPG)	Reserved (IPG)
		22	Reserved (IPG)	Reserved (IPG)
		23	Reserved (MNav)	Reserved (MNav)
		24	Normal Operation	Low Temperature
		25	Normal Operation	Power Supply Alarm
		26	Normal Operation	GND Leakage
		27	Normal Operation	Module Error
		28	Normal Operation	Aiming Beam Alarm
		29	Normal Operation	Critical Error
		30	Interlocks OK	1 or more interlocks are open
		31	Reserved (IPG)	Reserved (IPG)
SWB	Selects the working beam to read from or activate/deactivate.	Beam ID	Description	Sent: “SWB 1” Response: “SWB 1”
		0	970 nm beam	
		1	1064 nm beam	
PSOFF	Set the system into Standby Mode (diode power supply is turned off)			Sent: “PSOFF” Response: “PSOFF”
SFLT	Set the system into Fault Mode (diode power supply off)			Sent: “SFLT” Response: “SFLT”

TROUBLESHOOTING

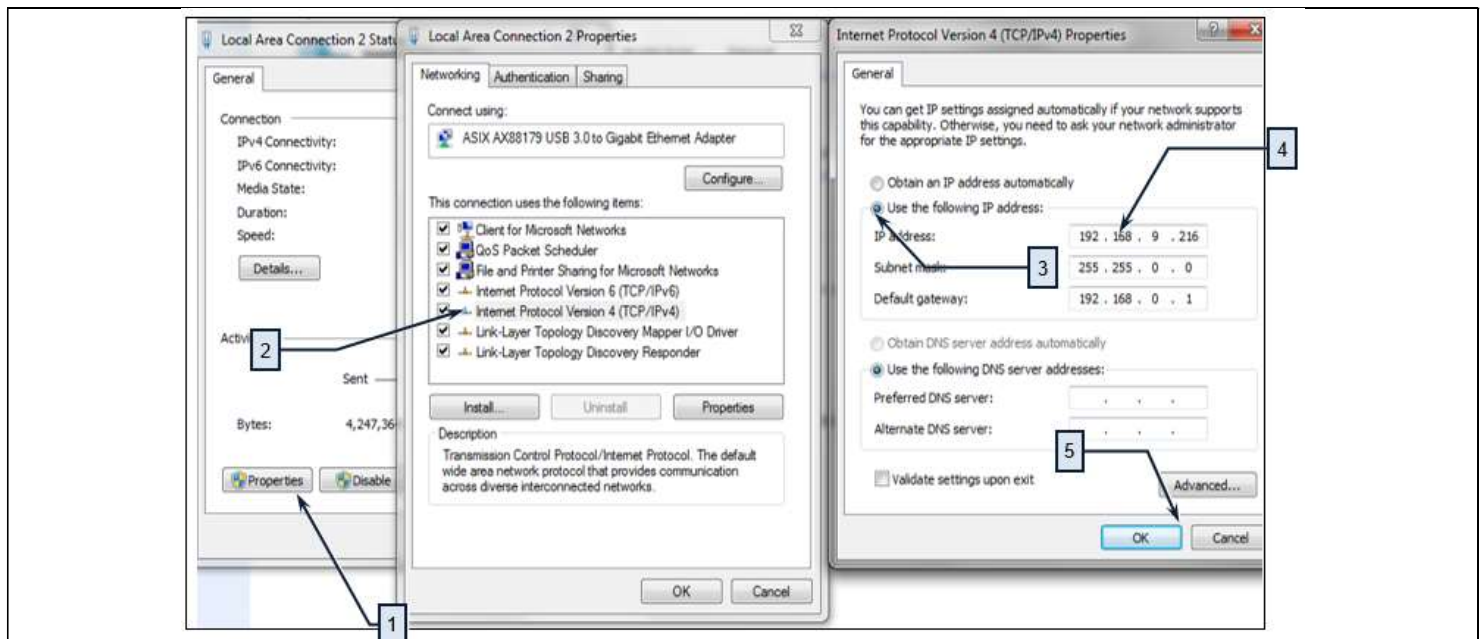
Computer Configuration Procedure

Ethernet Configuration Option

⚠ The following procedure assumes that the user is already familiar with operating their respective laser. In addition, it is strongly recommended that you review all safety and operational procedures before proceeding.

1. Connect PC to the Laser using standard Ethernet cable.
2. Manually configure the Local Area Connection settings for Ethernet (refer to Figure 6: Configuring the Local Area Connection for Ethernet).
3. Open your web browser and enter the IP address of this Internet Protocol.
4. If the connection is successful, the -Web User Utility web page will appear.

Figure 6: Configuring the Local Area Connection for Ethernet



STEP	DESCRIPTION
1	Navigate to the Local Area Connection screen on your computer and click the “Properties” command button.
2	Select the Internet Protocol TCP/IP Setting (TCP/IP 4 on Windows 7 OS) by double-clicking or click the Properties command button..
3	Click the Radio Button to “Use the following IP address” to manually assign the IP address.
4	Assign the IP address to 192.168.59.7x (x can not be 0 as it is the default IP address of the Laser). Assign the Subnetmask to 255.255.0.0 by clicking on the Textbox.
5	Pres the “OK” command button to accept these manual changes.

SERVICE

Service and Repairs

There are no operator serviceable parts inside. Please refer all internal servicing to qualified IPG personnel.

Many issues and questions regarding the safety, set-up, operation and maintenance of the IPG products can be resolved by reading this User's Guide carefully. However, if you have questions regarding the safety, set-up, operation or maintenance of your IPG product, please call the Customer Service department. If you cannot resolve the issues by using this User's Guide or over the telephone with our technical support group, you may need to return the product to IPG.

Serviceable Items

⚠ WARNING: The unit must never be operated with any of the covers removed.

The input voltage to the laser is potentially lethal. All electrical cables and connections should be treated as if it were a harmful level. All parts of the electrical cable, connector or device housing should be considered dangerous.

This device is classified as a high power **Class 4** laser instrument under 21 CFR 1040.10. This product emits invisible laser radiation at two independent channels. One channel emits invisible laser radiation at or around a wavelength of **975nm**, and the total light power radiated from the optical output is greater than **18 watts CW**. The second channel emits invisible laser radiation at or around a wavelength of **1064 nm**, and the total light power radiated from the optical output is greater than **46 watts CW**. The product is also equipped with a Class 2M visible laser operating at a wavelength at or around **658nm** and the total light radiated from this visible laser is **<1mW CW**. Laser emission of each channel and the visible laser is delivered via a single fiber optic delivery cable terminated with an SMA connector.

⚠ WARNING: Service personnel should always follow correct Lockout/Tagout procedures per your company's policy to ensure all potential energy is removed from the system before servicing.

Optical SMA Inspection and Cleaning Guide

⚠ CAUTION: It is imperative that the fiber SMA is checked for dust, dirt, or damage every time before it is connected with an optical head or beam coupler. The use of a dirty, or improperly cleaned, SMA can lead to serious damage to the laser. IPG Photonics is not responsible for any damages due to contaminated connectors. Tampering with the SMA without training by IPG will void the warranty.

For cleaning a SMA, you need the following materials:

- Powder free rubber gloves or finger cots
- Automatic connector cleaner (CLETOP)
- Optical Wipe
- Isopropanol (optical grade, water free)
- Compressed/canned air (oil free, water free)

IMPORTANT: It is imperative that you wear powder free rubber gloves during this cleaning procedure.

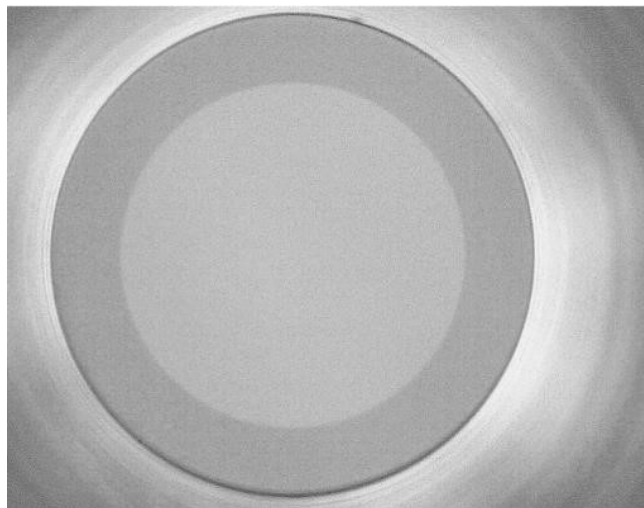
It is hereby stated that damage to the SMA can occur due to mishandling, the use of incorrect cleaning procedures or chemicals for cleaning and is not covered by the warranty.

Check that the SMA is free of any contamination. If the SMA is contaminated with any debris, it must be cleaned before operating the laser.

Refer to each solvent's MSDS (Material Safety Data Sheet) for additional information.

1. Turn the Main Power Supply off and disconnect the power cabling to the laser.
2. Remove the SMA cap and inspect the SMA end face via camera/microscope.
 - If the SMA is free of dirt, grease or small particles, no further cleaning is necessary.
 - Use canned/compressed air to remove loose particles
3. If the SMA was cleaned with canned/compressed air, inspect the SMA after cleaning to ensure a clean surface. Figure 7 shows a good quality SMA end face.
4. If the surface is perfectly clean, than place the SMA in the proper holder. Repeat if, not cleaned appropriately.

Figure 7: Example of a clean SMA end face



WARRANTY

Limited Express Product Warranties

IPG warrants to Buyer or, if Buyer is an authorized IPG reseller or distributor, to Buyer's original customer of the Product or Service, that Products delivered hereunder which are standard products of IPG will conform to their applicable specifications and be free from defects in materials and workmanship, and that Services provided by IPG will be performed in a workmanlike manner. For goods which are not standard products of IPG, such as developmental or custom designed goods, IPG warrants to Buyer that such goods delivered hereunder will conform to their applicable specifications and be free of defects in materials and workmanship upon receipt by Buyer. These non-transferable warranties start on the shipment date from IPG (or other date specifically referencing the warranty start date in IPG's sales order/order acknowledgement), and continue until the end of the warranty period listed in IPG's sales order/order acknowledgement. If there is no warranty period listed, then warranty period is one year. Products or major components manufactured by parties other than IPG bear the original manufacturer's warranty and warranty period. Buyer's sole and exclusive remedy, and IPG's exclusive obligation and liability, with respect to IPG's warranties is, at IPG's sole option, (i) for Product, to repair or replace the affected Product and correct the deficiencies and (ii) for Services, for IPG to re-perform the affected Services. IPG warrants repaired or replaced Products under warranty only for the remaining un-expired period of time in the original warranty. IPG reserves the right to issue a credit note for any defective Products that have proved defective through normal usage; Buyer debit memos are not allowed. This warranty governs over any conflicting terms in Buyer's purchase order or other IPG documents except as expressly provided herein..

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2.5 For the rights granted in this Agreement, Customer shall pay to IPG the price for the IPG hardware product in which the Licensed Software is embedded.

2.6 You understand that IPG may, at any time and in its sole discretion, update or modify the Licensed Product or discontinue updating and/or supporting the Licensed Product. In the case of an update or modification, you agree to allow IPG to automatically install the Licensed Software on the IPG hardware product on which the Licensed Software is used. Any updated or modified Licensed Product made available and/or installed by IPG on the IPG hardware product shall become part of the Licensed Software and subject to this Agreement.

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3.1 You may terminate the license granted hereunder at any time by destroying the Licensed Product together with all copies thereof and notifying IPG in writing that all use of the Licensed Product has ceased and that same has been destroyed.

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PRODUCT RETURNS

Returns to the United States

Note: All product returns require a Return Merchandise Authorization (RMA) from IPG.

To obtain an RMA, call the Customer Service department of IPG Photonics Corporation at 508-373-1100 (US) or +49 2736 44 20 451 (Germany).

If you return a product with a RMA, please perform the following procedure:

1. Products must be carefully packed in a suitable shipping container(s). Buyer assumes all responsibility for products damaged in shipment to IPG.
2. Buyer must issue a purchase order for the value of the replaced parts/service items and IPG will issue credit or invoice when the parts/service is received. Speak to IPG Service Manager for the amount authorized under the required purchase order.
3. All requests for repair or replacement under this warranty must be made to IPG within 30 days after discovery of the defect (but not later than 7 days after warranty expiration).
4. All products returned to IPG but which meet applicable specifications, not defectively manufactured or used not in accordance with this User's Guide, will result in the Buyer being charged IPG's standard examination charge.
5. Complete packing list with product model and serial number will ensure prompt repair.
6. Be sure to include with the returned product your 'ship to' address for the return of the serviced product.

Shipping Instructions:

Warranty Returns - Domestic & International Buyers* pay for one-way freight costs and insurance to IPG. IPG will pay for freight return cost and insurance back to the Buyer.

Non-Warranty Returns - Domestic & International Buyers* pay for two-way freight costs and insurance to IPG. If shipment consists of returns that are both warranty and non-warranty, the shipment will be considered as non-warranty.

Shipping address for returns to US:

*IPG Photonics Corporation
50 Old Webster Road
Oxford, MA 01540
Attn: Product Returns
Tel: 508-373-1100*

* International Returns must include applicable DUTIES AND TAXES, and you must mark air bills with "U.S. GOODS, RETURNED FOR REPAIR."

Returns to Germany

Shipping address for returns to Germany:

IPG Laser GmbH
Siemensstrasse 7
D-57299 Burbach, Germany
Attn: Product Returns
Tel: +49-(0)2736-44-20-451

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6. All returns must be packaged adequately to avoid damage during shipment.
7. Complete packing list with product model and serial number will insure prompt repair, if the other terms of this form are followed.
8. See the IPG Terms and Conditions for the applicable warranty for the products before you request the return of the products.
9. RMA number will expire 31 days after the date of issue. Thereafter, units received in under the expired RMA number will result in a longer turnaround time. Please include a copy of the completed RMA form with the return of your unit(s).

GLOSSARY

°C	Degrees centigrade or Celsius
°F	Degrees Fahrenheit
λ	Lambda (wavelength symbol)
μ s	Microsecond = 10^{-6} second
Amp	Amperes
AC	Alternating current
ADC	Analog-to-digital converter
ASCII	American Standard Code for Information Interchange (U.S. Government)
BTU	British thermal unit
CDRH	Center for Devices and Radiological Health (U.S. Government)
CFR	Code of Federal Regulation (U.S. Government)
cm	Centimeters = 10^{-2} meters
CPU	Central processing unit
CW	Continuous wave (operating mode)
DC	Direct current
EN	European Norm
GND	Ground
Hz	Hertz or cycles per second (frequency)
kg	Kilograms
kV	Kilovolts = 10^3 volts
kW	Kilowatts = 10^3 watts
l	Liters (volume)
lbs	Pounds
IP	Internet protocol
LD	Laser diode
LCD	Liquid crystal display
LED	Light emitting diode
nm	Nanometer = 10^{-9} meters
mA	Milliamps = 10^{-3} amperes
mm	Millimeter = 10^{-3} meters
ms	Millisecond = 10^{-3} seconds
MHz	Megahertz = 10^6 Hertz
mrad	Milliradian = 10^{-3} radians (geometry)
Pbr	Power (back reflection)
rms	Root mean square or quadratic mean
QCW	Quasi-Continuous wave (operating mode)
TCP	Transmission control protocol
VAC	Voltage alternating current
W	Watts (power)